Homework # 1

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Course: Bioinformatics Pipelines

1. Nextflow installation

   Nextflow was installed using conda environment

   $conda create -n nextflow_env -c bioconda

2. Runnin rnaseq-nf workflow with test data:

   with Docker

```
(nextflow_env) pavel@pavel-Lenovo-IdeaPad-S340-15API:~/BioinfPipelines/HW1/rnaseq-nf$ nextflow run nextflow-io/rnaseq-nf -profile docker

 N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [insane_newton] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/pavel/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/pavel/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[44/a3a3cf] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[62/b2fba2] process > RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[4b/70a4fa] process > RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[4e/f6b77c] process > MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 22:44:52
Duration    : 4m 27s
CPU hours   : 0.1
Succeeded   : 4


(nextflow_env) pavel@pavel-Lenovo-IdeaPad-S340-15API:~/BioinfPipelines/HW1/rnaseq-nf$
```

   with Conda

```
(nextflow_env) pavel@pavel-Lenovo-IdeaPad-S340-15API:~/BioinfPipelines/HW1/rnaseq-nf$ nextflow run nextflow-io/rnaseq-nf -profile conda

 N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [insane_colden] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/pavel/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/pavel/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[d9/ac57bf] process > RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[93/49c466] process > RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[3a/580e11] process > RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[bf/787bdb] process > MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 22:51:23
Duration    : 5m 39s
CPU hours   : (a few seconds)
Succeeded   : 4
```

3. Running docker with some dataset from NCBI

SRR37943487 was obtained partially using SRA-toolkit:

$fasterq-dump -X 10000 --split-files SRR37943487

```
(nextflow_env) pavel@pavel-Lenovo-IdeaPad-S340-15API:~/BioinfPipelines/HW1/rnaseq-nf/from_NCBI$ nextflow run nextflow-io/rnaseq-nf   --reads '/home/pavel/BioinfPipelines/HW1/rnaseq-nf/
from_NCBI/SRR37943487_{1,2}.fastq.gz'   --transcriptome '/home/pavel/BioinfPipelines/HW1/rnaseq-nf/from_NCBI/Homo_sapiens.GRCh38.cdna.all.fa'   --outdir 'my_results'   -profile docker


 N E X T F L O W   ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [determined_torricelli] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/pavel/BioinfPipelines/HW1/rnaseq-nf/from_NCBI/Homo_sapiens.GRCh38.cdna.all.fa
reads        : /home/pavel/BioinfPipelines/HW1/rnaseq-nf/from_NCBI/SRR37943487_{1,2}.fastq.gz
outdir       : my_results

executor >  local (4)
[b8/aca479] process > RNASEQ:INDEX (Homo_sapiens) [100%] 1 of 1 ✔
[87/846622] process > RNASEQ:FASTQC (SRR37943487) [100%] 1 of 1 ✔
[ff/f7613f] process > RNASEQ:QUANT (SRR37943487)  [100%] 1 of 1 ✔
[89/81caf5] process > MULTIQC                     [100%] 1 of 1 ✔

Done! Open the following report in your browser --> my_results/multiqc_report.html

Completed at: 10-Apr-2026 00:58:29
Duration    : 18m 46s
CPU hours   : 0.3
Succeeded   : 4
```